STAFF REVIEW · H446 181 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Do Now — source answers

1. It sends the value 8 back to the caller. A return statement also ends the current invocation.

2. 1, 2 and 6. Each iteration multiplies the previous total by the new i.

3. C. A stack is last in, first out (LIFO).

4. The inner AddTwo(3) returns 5; then the outer AddTwo(5) returns 7.

I Do — source worked steps

1. Calls — 3 → 2 → 1 → 0. The argument decreases by one each time.

2. Base case — Factorial(0) returns 1. No more calls occur.

3. Returns — Factorial(1)=1, Factorial(2)=2, Factorial(3)=6. Final answer: 6.

We Do — guided CHECK

1. Predict the first return
Factorial(0) returns 1.
Complete the return chain
Factorial(1)=1, Factorial(2)=2, Factorial(3)=6, Factorial(4)=24.

You Do — source mark schemes

1. Calls: 2 → 1 → 0. Return from 0 is 1, then 1 returns 1 and 2 returns 2.

2. Calls continue without a stopping condition, adding stack frames until the runtime eventually fails (typically stack overflow).

3. Before the base case: if (n < 0) throw new ArgumentOutOfRangeException(nameof(n));. Negative values would otherwise keep decreasing and never reach 0.

Exit Ticket — source answer

1. False. Previous calls are still waiting; they return as the stack unwinds.